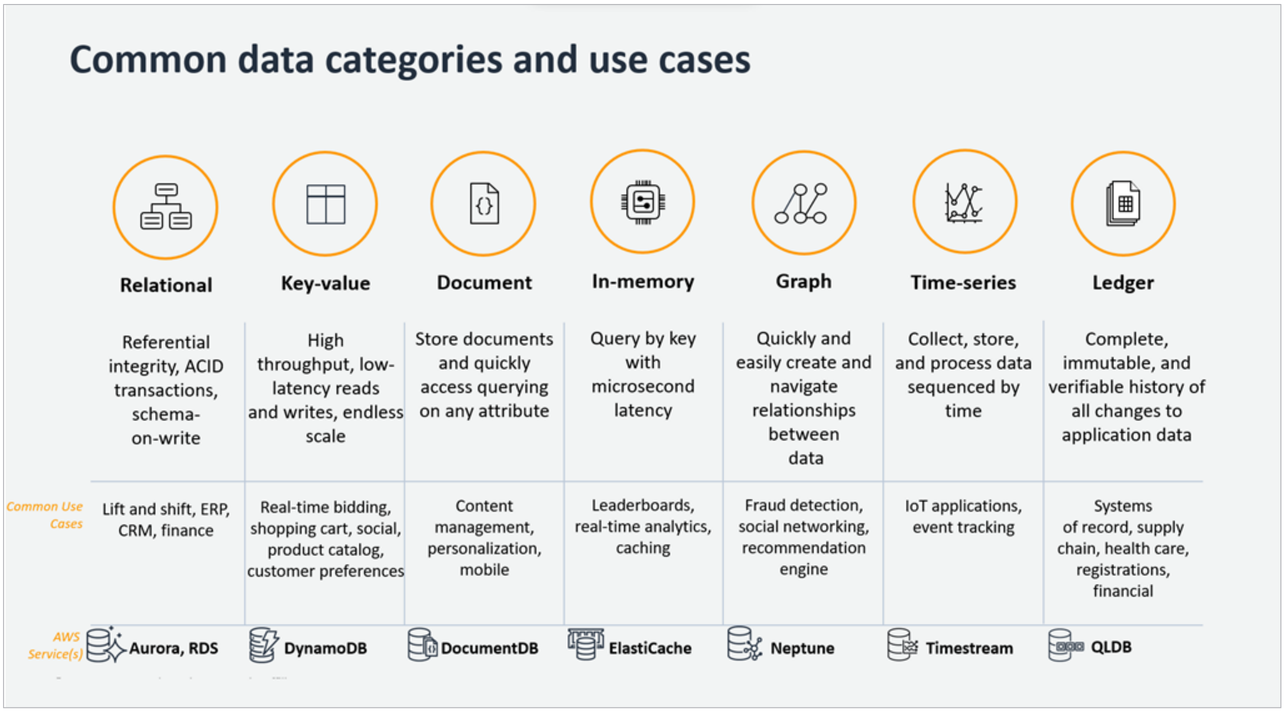

Figure 3: Data category and use cases

Airline Use Case	Recommended AWS Purpose-Built Database
Increased Operational Inefficiency	Amazon Redshift Amazon
Enhanced Customer Experience	DynamoDB
Data driven Decision-making	Amazon Aurora
Effective Risk Management	Amazon Neptune
Accurate Forecasting and Planning	Amazon RDS (Relational Database Service)

Figure 4: Purpose-built database for our use case

tasks such as hardware provisioning, software patching, and database backups allowing organisations to focus on their core competencies.

Purpose-built database services offered by AWS

AWS offers a wide range of purpose-built database services that cater to specific use cases and requirements. These services are designed to provide organisations with scalable, high-performance, secure, and cost-effective database solutions.

The first factor to consider when choosing a purpose-built database is the application workload, which falls into three categories: transactional (OLTP), analytical (OLAP), and caching. Transactional workloads involve high concurrency with small data operations, while analytical workloads aggregate large volumes of data for reporting. Caching workloads store frequently accessed data for faster response times. Understanding the workload helps in selecting the right database for a specific use case.

The second factor to consider when choosing a purpose-built database is the shape of data. This includes understanding the entities, relationships, access patterns, and update frequency. Common data models include relational, key-value or wide-column, document, and graph. Assessing a data's shape helps determine the best-suited database for that application.

When choosing a purpose-built database, it's also important to consider the performance requirements of the application. Factors to assess include data access speed, record size, and geographic considerations. For critical workloads with users expecting quick responses, an in-memory cache can help reduce latency. If the focus is on internal analytics or background data processing, handling large data volumes becomes a priority. Some databases offer data replication across multiple regions, minimising response times by keeping data closer to users. Taking these performance factors into account ensures the selected database aligns with the application's specific needs.

Finally, when choosing a purpose-built database, consider the operations burden. AWS purpose-built databases handle many operations automatically, including instance failures, backups, and upgrades. With fully managed databases from AWS, you can focus on developing features and innovating for